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COMMON PREPERIODIC POINTS FOR QUADRATIC POLYNOMIALS

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ABSTRACT. Let $f_c(z) = z^2 + c$ for $c \in \mathbb{C}$. We show there exists a uniform upper bound on the number of points in $\mathbb{P}^1(\mathbb{C})$ that can be preperiodic for both f_{c_1} and f_{c_2}, for any pair $c_1 \neq c_2$ in $\mathbb{C}$. The proof combines arithmetic ingredients with complex-analytic: we estimate an adelic energy pairing when the parameters lie in $\overline{\mathbb{Q}}$, building on the quantitative arithmetic equidistribution theorem of Favre and Rivera-Letelier, and we use distortion theorems in complex analysis to control the size of the intersection of distinct Julia sets. The proofs are effective, and we provide explicit constants for each of the results.

1. INTRODUCTION

Consider the family of quadratic polynomials

$$f_c(z) = z^2 + c$$

for $c \in \mathbb{C}$, each viewed as a dynamical system $f_c : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ on the Riemann sphere. Recall that a point $z \in \hat{\mathbb{C}}$ is said to be *preperiodic* if its forward orbit under f_c is finite. It is well known that the set of all preperiodic points for f_c will determine c. Indeed, we have

$$\text{Preper}(f_{c_1}) = \text{Preper}(f_{c_2}) \iff J(f_{c_1}) = J(f_{c_2}) \iff c_1 = c_2 \tag{1.1}$$

in this family, where $J(f_c)$ is the Julia set and $\text{Preper}(f_c)$ is the set of preperiodic points [2]; see §2.3 for more information.

For any $c_1 \neq c_2$ in $\mathbb{C}$, the intersection of $\text{Preper}(f_{c_1})$ and $\text{Preper}(f_{c_2})$ is finite ([3, Corollary 1.3] and [31, Theorem 1.3]), even though their closures, which contain the Julia sets of f_{c_1} and f_{c_2}, respectively, can have complicated, infinite intersection. We investigate the question of how many preperiodic points are required to uniquely determine the polynomial, without the information of the period or length of an orbit. We prove:

THEOREM 1.1. *There exists a uniform constant B so that*

$$|\text{Preper}(f_{c_1}) \cap \text{Preper}(f_{c_2})| \leq B$$

for every $c_1 \neq c_2$ in $\mathbb{C}$.

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where Diag is the diagonal, assuming $\log|z-w|$ is in $L^1(\rho\otimes\rho')$. If the measures have total mass 0 with continuous potentials on $\mathbb{P}^1$, we have $(\rho,\rho)\geq 0$ with equality if and only if $\rho=0$. Similarly, one defines

$$(\rho,\rho')_v := -\iint_{\mathbf{A}^1_v\times\mathbf{A}^1_v\smallsetminus\mathrm{Diag}} \delta_v(z,w)\,d\rho(z)\,d\rho'(w) \tag{9.1}$$

on the Berkovich line over $\mathbb{C}_v$, with respect to a non-archimedean valuation, where $\delta_v(z,w)$ is the logarithm of the Hsia kernel in place of $\log|z-w|_v$. See [5, Proposition 4.1] and further information throughout Chapters 4 and 5 of [5].

Now let K be a number field. An adelic measure is a collection $\mu=\{\mu_v\}_{v\in M_K}$ of probability measures on the Berkovich $\mathbf{P}^{1,an}_v$, with continuous potentials at all places v and for which all but finitely many are trivial (meaning that they are supported at the Gauss point). For any adelic measure μ, a height function is defined on $\mathbb{P}^1(\overline{\mathbb{Q}})$ by

$$h_\mu(F) := \frac{1}{2}\sum_{v\in M_K}\frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\,([F]-\mu_v,[F]-\mu_v)_v,$$

where F is any finite, $\mathrm{Gal}(\overline{K}/K)$-invariant subset of $\overline{K}$, and $[F]$ is the probability measure supported equally on the elements of F. To define h_μ at all points of $\mathbb{P}^1(\overline{\mathbb{Q}})$, we put

$$h_\mu(\infty) := \frac{1}{2}\sum_{v}\frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}(\mu_v,\mu_v)_v.$$

The equidistribution theorems of [19, 4, 12] state that if F_n is a seqence of $\mathrm{Gal}(\overline{K}/K)$-invariant finite sets with $h_\mu(F_n)\to 0$ and $|F_n|\to\infty$ as $n\to\infty$, the discrete probability measures

$$[F_n] := \frac{1}{|F_n|}\sum_{x\in F_n}\delta_x$$

converge weakly to the measure μ_v on $\mathbf{P}^{1,an}_v$ at each place v of K.

There is a pairing between any two such heights, h_μ and h_ν, associated to adelic measures μ and ν, as

$$\langle h_\mu,h_\nu\rangle = \frac{1}{2}\sum_{v\in M_K}\frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}(\mu_v-\nu_v,\mu_v-\nu_v)_v. \tag{9.2}$$

It satisfies $\langle h_\mu,h_\nu\rangle=0 \iff h_\mu=h_\nu \iff \mu=\nu$. The energy pairing (1.3) between two quadratic polynomials is a special case, taking the dynamical canonical heights $\hat{h}_{c_1}$ and $\hat{h}_{c_2}$ associated to their adelic equilibrium measures.

REMARK 9.1. The height h_μ is defined for an arbitrary adelic measure, but small sequences (meaning the sequences $\{F_n\}$ of Galois-invariant sets with $h_\mu(F_n)\to 0$ and $|F_n|\to\infty$) do not always exist.

9.2. **Height pairing as a distance.** Following [20], we consider a distance between two adelic measures $\mu = \{\mu_v\}$ and $\nu = \{\nu_v\}$ on $\mathbb{P}^1$ over a number field K, defined by

$$d(\mu,\nu) := \langle h_\mu, h_\nu\rangle^{1/2},$$

where $\langle h_\mu, h_\nu\rangle$ was defined in (9.2); see [20, Theorem 1].

Suppose that c_1 and c_2 are elements of a number field K. Let $\mu_1 := \{\mu_{c_1,v}\}_{v\in M_K}$ and $\mu_2 := \{\mu_{c_2,v}\}_{v\in M_K}$ be the equilibrium measures of f_{c_1} and f_{c_2}, respectively. Let F be any finite, nonempty, $\mathrm{Gal}(\overline{K}/K)$-invariant subset of $\mathbb{P}^1(\overline{\mathbb{Q}})$. Let $[F]$ denote the probability measure supported equally on the elements of F. For each place v of K, choose a positive real $\varepsilon_v > 0$, with $\varepsilon_v = 1$ for all but finitely many v. The collection $\varepsilon := \{\varepsilon_v\}_{v\in M_K}$ will be called an adelic radius. As in [19], we consider the adelic measure $[F]_\varepsilon$, defined as a regularization of the probability measure $[F]$: it is supported on the circles of radius ε_v about each point of F. At a non-archimedean place, this means the Type II or III point associated to the disk of radius ε_v. The triangle inequality implies that

$$\langle f_{c_1}, f_{c_2}\rangle^{1/2} = d(\mu_1,\mu_2) \le d(\mu_1,[F]_\varepsilon) + d(\mu_2,[F]_\varepsilon) \tag{9.3}$$

for any choices of F and ε.

It is worth noting that, if the radius $\varepsilon_v \to 0$ at some place, then the right-hand-side of (9.3) will tend to ∞. This is because the potential of the measure $[F]_\varepsilon$ at v will blow up near the points of F. On the other hand, for ε_v too large, the measure $[F]_\varepsilon$ is not a good approximation of $[F]$. Thus, for (9.3) to be useful in our proof of Theorem 1.1, we will need to choose ε well. This general strategy also appears in the proofs of [19, Theorem 3] and in [20, Proposition 13]. In our case, the choice of $\varepsilon = \{\varepsilon_v\}_{v\in M_K}$ will be governed by Proposition 3.8 and its non-archimedean counterparts, and this leads to Theorem 1.8.

LEMMA 9.2. *Let K be a number field and fix $c_1 \neq c_2$ in K. We have*

$$\langle f_{c_1}, f_{c_2}\rangle^{1/2} \le \sum_{i=1}^{2}\left(\sum_{v\in M_K}\frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\left(-(\mu_i,[F]_\varepsilon)_v + \frac{\log(1/\varepsilon_v)}{2|F|}\right)\right)^{1/2}$$

for any choice of finite, non-empty, $\mathrm{Gal}(\overline{K}/K)$-invariant subset F of $\overline{\mathbb{Q}}$ and any adelic radius $\varepsilon = \{\varepsilon_v\}_{v\in M_K}$.

Proof. We first observe that

$$\begin{aligned} d(\mu_i,[F]_\varepsilon)^2 &= \frac{1}{2}\sum_v \frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\,(\mu_i - [F]_\varepsilon, \mu_i - [F]_\varepsilon)_v \\ &= \sum_v \frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\left(-(\mu_i,[F]_\varepsilon)_v + \frac{1}{2}([F]_\varepsilon,[F]_\varepsilon)_v\right) \end{aligned}$$

because $(\mu_i,\mu_i)_v = 0$ at every place. The self-pairing of $[F]_\varepsilon$ can be estimated in terms of the self-pairing of $[F]$ ([20, Lemma 12] and [19, Lemma 4.11]), as

$$([F]_\varepsilon,[F]_\varepsilon)_v \le ([F],[F])_v + \frac{\log(1/\varepsilon_v)}{|F|}.$$

But observe that

$$\sum_v \frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}([F],[F])_v = 0$$

by the product formula on K. So the triangle inequality (9.3) completes the proof of the proposition. □

9.3. **Proof of Theorem 1.8.** Fix any $L \geq 27$ and recall the definition of the auxiliary height h_L on $\mathbb{A}^2(\overline{\mathbb{Q}})$ from §8.1. An appropriate choice of $\varepsilon = \{\varepsilon_v\}$ in Lemma 9.2 gives

PROPOSITION 9.3. *Fix any $L \geq 27$. Fix c_1 and c_2 in $\overline{\mathbb{Q}}$, and assume f_{c_1} and f_{c_2} have $N > 1$ preperiodic points in common in $\mathbb{P}^1(\overline{\mathbb{Q}})$. Then for all $0 < \delta < 1/4$, we have*

$$\langle f_{c_1}, f_{c_2}\rangle \leq 4\left(\delta + \frac{3\log(1/\delta)}{2(N-1)}\right) h_L(c_1,c_2).$$

Proof. Fix a number field K containing c_1 and c_2. Let F be the $\mathrm{Gal}(\overline{K}/K)$-invariant set of common preperiodic points for f_{c_1} and f_{c_2} in $\overline{\mathbb{Q}}$, so that $|F| = N-1$. For each place $v \in M_K$, recall the definition of ℓ_v from §8.1. Fix $0 < \delta < 1/4$ and set

$$\varepsilon_v = \delta^{3\ell_v}.$$

Note that $\varepsilon_v = 1$ for all but finitely many places $v \in M_K$.

For each archimedean place v, note that

$$\varepsilon_v = e^{-3\ell_v \log(1/\delta)} = \max\{|c_1|_v, |c_2|_v, L\}^{-3\log(1/\delta)},$$

so Proposition 3.8 implies that

$$\lambda_{c_i,v}(z) \leq \delta\,\ell_v$$

for any point z within a neighborhood of radius ε_v of the filled Julia set K_{c_i}. As all points of F lie in K_{c_i}, this implies that

$$-(\mu_i,[F]_\varepsilon)_v \leq \delta\,\ell_v$$

for this choice of ε_v and each i.

Similarly for each non-archimedean place $v \nmid 2$, we apply Proposition 5.2, and for each non-archimedean $v \mid 2$, we apply Proposition 6.3.

Summing over all places, we find that

$$\begin{aligned}\sum_{v\in M_K} \frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\left(-(\mu_i,[F]_\varepsilon)_v + \frac{\log(1/\varepsilon_v)}{2|F|}\right) &\leq \sum_v \frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\left(\delta\,\ell_v + \frac{3\log(1/\delta)}{2|F|}\,\ell_v\right)\\ &= \left(\delta + \frac{3\log(1/\delta)}{2|F|}\right) h_L(c_1,c_2).\end{aligned}$$

Lemma 9.2 then implies

$$\begin{aligned}\langle f_{c_1}, f_{c_2}\rangle^{1/2} &\leq \sum_{i=1}^{2}\left(\sum_{v\in M_K} \frac{[K_v:\mathbb{Q}_v]}{[K:\mathbb{Q}]}\left(-(\mu_i,[F]_\varepsilon)_v + \frac{\log(1/\varepsilon_v)}{2|F|}\right)\right)^{1/2}\\ &\leq 2\left(\left(\delta + \frac{3\log(1/\delta)}{2|F|}\right) h_L(c_1,c_2)\right)^{1/2}.\end{aligned}$$